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Oefeningenreeks 5D nr.1.3

$f(x) = \sqrt[3]{x}$ op het interval $[0,8]$

$$f(x) = x^{\frac{1}{3}}$$

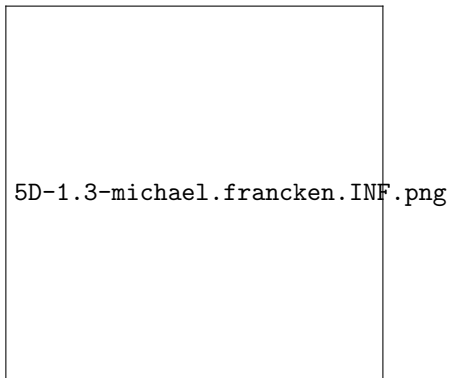
$$F(x) = \frac{x^{\frac{1}{3}+1}}{\frac{1}{3}+1}$$

$$= \frac{3}{4} x^{\frac{4}{3}}$$

$$= \left[\frac{3}{4} x^{\frac{4}{3}} \right]_0^8$$

$$= \left(\frac{3}{4} \cdot 8^{\frac{4}{3}} \right) - \left(\frac{3}{4} \cdot 0^{\frac{4}{3}} \right)$$

$$= \frac{3}{4 \sqrt[3]{8^4}} = 3 \cdot 4 = 12$$



References